

Benjamin Critoph

Software Developer | Computer Science Graduate | 3 Co-ops

Pickering, ON | ben.critoph.job@gmail.com | 416-988-8683 | [LinkedIn](#) | [Github](#) | [Portfolio](#)

Work Experience

Software Developer Intern | [Camis Inc](#)

Jan 2024 - Sep 2024

- Delivered **self-service checkout** functionality in **Angular/RxJS**, enabling campers to complete reservations on mobile devices without in-person checkout
- Developed and maintained features for a high-traffic campsite reservation platform serving **millions of users**, resolving critical bugs and improving booking workflows
- Worked within a **.NET/C#** reservation and **CMS** ecosystem managing campground configuration and business rules across **1,300+ parks and facilities**
- Enhanced the booking search interface with **customizable icons** and **touch-driven mobile navigation**, improving usability and configurability
- Contributed to **WCAG compliance** by resolving accessibility issues identified through **external audits**

Software Developer Intern | [Brock Solutions](#)

May 2022 - Sep 2022

- Automated data cleanup for a baggage sortation system upgrade at Kahului Airport using **Python** to fill missing data in large legacy Excel datasets and **SQL/SSMS** to prepare data for the upgraded system
- Customized and executed **Factory Acceptance Testing (FAT)** procedures to validate system behaviour and ensure operational requirements were met
- Diagnosed and resolved deployment issues** during system testing, supporting a successful rollout of critical airport infrastructure
- Presented FAT procedures** and test results directly to clients, explaining technical findings and requirements to non-development stakeholders

Junior Developer Intern | [Hatch Ltd](#)

Sep 2021 - Dec 2021

- Developed and enhanced internal enterprise web applications using **Angular, TypeScript, C#/.NET** and **NGRX** for state management
- Revamped an internal Estimate Governance application by **migrating authentication** from **ADAL** to **MSAL** and implementing **AG-Grid** to improve data visualization, sorting, and organization
- Integrated Checkmarx into **Azure DevOps CI/CD pipelines**, reducing manual code review effort by **~15%** and improving security compliance

Projects

Japanese Lyrics Learning App | [Live Demo](#) | [GitHub](#)

2026

- Built an interactive language-learning platform using **Next.js, TypeScript, React, and Supabase**, featuring synchronized playback and dynamic Kanji, Furigana, and Romaji lyric displays
- Built a **lyric processing pipeline** using **LRCLIB, Kuromoji, Jisho, and Google Translate APIs** to fetch synchronized lyrics, tokenize Japanese text, retrieve dictionary definitions, and generate line-by-line translations
- Implemented user vocabulary management and Anki export, allowing users to save words with definitions, conjugations, and lyric-line context for personalized study

Multiplayer Cross Cribbs Game | [Live Demo](#) | [GitHub](#)

2026

- Built a real-time multiplayer card game using **React, TypeScript, Node.js, and Socket.io**, supporting 1v1 and 2v2 gameplay with both local and online game modes
- Designed a **server-authoritative game architecture** with **Socket.io rooms** to synchronize game state across players, preventing client-side desynchronization and invalid game actions
- Developed a **responsive game interface** with round history, live scoring breakdowns, player hands, and game-state components, supporting desktop and mobile gameplay

Technologies and Languages

- Languages:** JavaScript/TypeScript, Python, Java, C#, C
- Frontend:** React, Angular, Next.js, HTML, CSS, Tailwind CSS, Material UI, RxJS, NGRX
- Backend:** Node.js, .NET
- Databases:** SQL, MongoDB, Firebase, Supabase
- Tools/Concepts:** Git, Azure DevOps, REST APIs, Socket.io, Agile, Responsive Design, CI/CD

Education

B.Comp. Computer Science (Co-op), University of Guelph

2019-2025